PROLOGUE

The Last Monopoly

THE CARDS

IN BROOKLYN IN THE late 1940s, a classics teacher named Alice Kober was trying to read a language that nobody on earth could read.

The language had been written down more than three thousand years earlier, on clay tablets found in the ruins of palaces on Crete and in mainland Greece. Archaeologists called the script Linear B. It had about ninety signs, too many for an alphabet and too few for a script of whole words, which suggested that each sign stood for a syllable. That was almost all anyone knew. Nobody knew what sounds the signs stood for, or what language they wrote. There was no dictionary, no bilingual inscription, nothing like the Rosetta Stone that had let scholars read Egyptian hieroglyphs. There were only the tablets: thousands of short lists of signs, pressed into clay by scribes who had been dead for three millennia.[1]

Kober did not try to guess what the words meant. She did something slower and stranger. She recorded, sign by sign, where each sign appeared and what appeared next to it. She copied the inscriptions onto cards and filed them by hand, year after year, until her index ran to well

over a hundred thousand items. Among her papers today are two of the cigarette cartons she used to store them.[2]

From those cards she noticed something nobody had proved before. Certain words appeared in small sets of closely related forms that differed in their endings, the way *house*, *houses* and *house's* differ in English. The language had endings. It changed the shapes of its words according to their role in a sentence. And because the endings changed in regular ways, Kober could begin to tell which signs probably shared a consonant and which shared a vowel, without knowing what any of those consonants or vowels were.[3]

She never learned whether she was right. Alice Kober died on 16 May 1950, at the age of forty-three.[4]

THE GRID

The work passed to a young English architect named Michael Ventris. As a schoolboy of fourteen, in 1936, he had met the archaeologist Arthur Evans, who had excavated the palace at Knossos and had spent decades failing to read its tablets. Ventris decided then that he would read them.[5]

He took Kober's approach and pushed it further. He drew up a grid: rows for signs he believed shared a consonant, columns for signs he believed shared a vowel. Every placement in the grid came from the same kind of evidence, which sign could replace which, which signs travelled together, which endings alternated with which. For a long time not one cell in the grid had a sound attached to it. It was a map of relations, and nothing else.[6]

Then, in early 1952, he made a guess. A few words appeared only on tablets from Crete. Perhaps, he thought, they were the names of Cretan towns, and towns keep their names for a very long time. If one of those words was *Amnisos*, a harbour near Knossos, then two of its signs had sounds. And once two signs had sounds, the grid told him the sounds of

others, because it already knew which signs shared what. He tried the new values on another word. It came out as *ko-no-so*. Knossos.[7]

Ventris had expected the language to be Etruscan, or something like it. As the values spread across the grid, more and more words came out looking like an early form of Greek. On 1 July 1952 he said so, on BBC radio. The tablets, he told his listeners, "must, after all, be written in Greek."[8]

THE TRIPOD

Plenty of experts were unconvinced. A grid of guesses can be made to say almost anything, if you want it to.

Then came a tablet that the grid had never seen. It had been dug up at Pylos, in southern Greece, in June 1952, and could be read only after it was cleaned the following spring. Next to its words, the scribe had drawn little pictures of the objects he was listing. The American archaeologist Carl Blegen, who had excavated it, read the words using Ventris's values. Beside a drawing of a three-legged cauldron was the word *ti-ri-po-de*: tripods. Beside a jar with four handles was a word meaning four-handled. Beside a jar with no handles, a word meaning without handles. Blegen wrote to Ventris. "All this seems too good to be true," he said.[9]

It was true. The relations among the signs had held the language all along. Kober and Ventris had found it without a dictionary, without a single translated word to start from, by studying how the signs leaned on one another. But it was the tripod, a drawing of a real thing standing next to a word, that told them they were right.

Michael Ventris died in a car crash on 6 September 1956. He was thirty-four.[10]

The story of the cards and the grid is, in miniature, the story this book tells. Some seventy years later a machine did something like what Kober and Ventris did, on a scale they could not have imagined: it worked through almost everything human beings have ever written

down, without being told what a single word meant. And like them, when it was done, it still needed something outside the words to tell it whether it was right.

THE OLDEST MONOPOLY

Many creatures communicate. A honeybee returning to the hive dances a small figure-of-eight that tells its sisters which way to fly and how far. Vervet monkeys in East Africa have one alarm call for leopards, another for eagles and a third for snakes.[11] Both systems work, but they are closed. The bee cannot dance about yesterday. The monkey cannot warn about a leopard that might come next week, or discuss whether leopards are, on the whole, a bad thing.

Human language is different in kind. With a few thousand words and a handful of patterns, any speaker can say something no one has ever said before, about things that are not here, that happened long ago or may never happen at all. Nobody knows how people first came to speak, and the question may have no answer, because speech leaves no trace. What we do know is that as far back as any record reaches, people have talked, and that every human community ever found has had a full language, as rich and as orderly as any other.[12]

Language let strangers cooperate. It let a grandmother pass on to her grandchildren what she had learned in seventy years, so that each generation began a little ahead of the last. Written down, it became law, scripture, accounting, science and poetry. Almost everything we call civilization is made of words.

And words were ours alone. Machines could lift heavier loads than us, run faster, add up columns of figures more quickly. That was fine. Those were the jobs of the body, or of the lower floors of the mind. Language was the thing that made us us.

DESCARTES' BET

In 1637 René Descartes proposed a way to tell a man from a machine. A machine, he wrote, might be built to utter words, even to cry out that it was hurt when touched in a certain place. But it could never arrange its words to answer the sense of whatever was said in its presence, as even the dullest of men can do.[13]

In 1950 the English mathematician Alan Turing turned Descartes' bet into a game. A judge types questions to two hidden players, one a person and one a machine, and tries to tell which is which. Turing guessed that by the end of the century a machine would fool the average judge a fair part of the time.[14] He was wrong about the date. But he had moved the question, almost without anyone noticing, from the machine to the person judging it.

For almost four centuries this looked like one of the safest bets in philosophy. When computers arrived, engineers spent decades trying to teach them language the obvious way: by writing it down. They wrote rules of grammar, rules of meaning, dictionaries of words and their definitions. They promised success within a few years. It never came. The next chapters will show why: meaning would not sit still long enough to be written down as rules.

THE THIRTIETH OF NOVEMBER

Then, on 30 November 2022, a small American company called OpenAI put a new program on the internet.

Nobody at OpenAI expected much. The launch was called a "research preview", and the blog post was modest.[15] The program was called ChatGPT, which is not the kind of name you give to something you expect to change the world. Five days later a million people had used it.[16]

You could ask it to explain the causes of the First World War, and then to explain them again for a ten-year-old. You could ask it for a sonnet about your cat, a cover letter for a job you were not qualified for,

or a recipe for the three things left in your fridge. You could tell it that it was wrong, and it would apologize, sometimes when it had been right. It invented facts with complete confidence. But whatever it was doing, it was answering the sense of what was said.

Descartes had lost his bet.

In less than four years, more than a billion people were using ChatGPT every week, and hundreds of millions more were using its rivals.[17] When economists looked at a large sample of real conversations, they found that about half the messages asked to *know* something: a fact, an explanation, a piece of advice. About two in five asked the machine to *do* something: write, rewrite, summarize, code. And about one in ten simply *expressed* something, the way you might to a friend: a worry, a feeling, a thought out loud. Most of these conversations had nothing to do with work.[18]

Were the people typing talking to a mind, or to something that only sounded like one? On this the experts split. A group of linguists and computer scientists had argued, the year before ChatGPT appeared, that such programs are "stochastic parrots": they stitch together words that tend to follow one another, with the form of language and none of its meaning.[19] Others look at what these systems now do, from olympiad mathematics to finding flaws in software that experts had missed, and conclude that "parrot" no longer describes it.[20] This book will give you what you need to judge for yourself. The place to begin is with how these machines were made.

NOBODY WROTE IT DOWN

When people imagine how a machine might learn to speak, they imagine something like the efforts of the 1960s. Somebody would have to teach it grammar, give it a dictionary, and explain to it what words mean and how the world works.

None of that happened.

The programs that broke Descartes' bet were built around a single, almost childish task. They were shown a piece of text with the next word hidden, and asked to guess it. *The cat sat on the* … Mat? Sofa? Roof? After each guess the program was shown the right answer and adjusted itself, very slightly, to guess better next time. Then it was shown another piece of text. And another. Trillions of times, across a large share of everything humans have ever written.[21]

Nobody gave it a rule of grammar or a list of facts, or told it that Paris is in France, that glass breaks when it falls, or that people who lose their jobs are often afraid. And yet, at the end, it could tell you all of these things, and explain them, and use them.

So where was the knowledge?

It was in no particular sentence. Take something everyone knows: that bananas are yellow. Almost nobody writes that down. It is too obvious to mention, and people write about what surprises them.[22] And yet a machine trained only on text knows it. It knows it because "banana" keeps company with "peel", "ripe", "sweet", "market" and "yellow" in a hundred thousand sentences about breakfasts, shopping lists, children's books and recipes, none of which is *about* the colour of bananas. The fact was never stated. It was spread out in the pattern of how words lean on one another, the way the Greek language had been spread out, unread, in the pattern of how Kober's signs leaned on one another.

That is the idea at the heart of this book. **Human knowledge, and much of human thinking, is stored in our language not explicitly but implicitly: as meaning that lives in the relations between words.** No single sentence contains it. It lives in the pattern of countless sentences: what goes next to what, what can stand in for what, what tends to follow from what.[23]

Think of a subway map. It bends the distances, and it leaves out the streets and the buildings.[24] What it keeps faithfully is what connects to what, and where the lines cross. And yet with that map alone, you can find your way across a city you have never seen. Ventris's grid was a map of that kind, for one language. Our writing, taken all together, is a

map of that kind for everything we know and much of how we reason. Nobody drew it. It accumulated, one sentence at a time, from everyone who ever wrote. And a machine that learned to guess the next word was forced, in order to guess well, to learn the map.

WHAT IS NOT BETWEEN THE WORDS

The subway map breaks in one important place. Whoever holds a subway map knows that there is a city outside. The machine has only the map.

About forty years ago, two psychologists at the University of Pennsylvania, Barbara Landau and Lila Gleitman, studied a little girl who had been blind from birth. They called her Kelli. At about three years old, Kelli used the words *look* and *see* correctly, though she had never seen anything. When the researchers blindfolded sighted children of the same age and asked them to "look up", the children tipped their faces toward the ceiling. When they asked Kelli, she kept her head still and reached up, searching the space above her with her hands. Told to touch a table but not to look at it, she tapped it. Told that now she could look at it, she explored its whole surface with her fingers. For Kelli, *look* meant to explore with the hands. She had worked out its place among the other words, its relations, with great precision, and filled it with the only kind of looking she had.[25]

Where had she learned this? Not from the situations she was in, the researchers found, but largely from how the words were used in the sentences around her: which words *look* and *see* kept company with, and in what kind of sentences. She was finding meaning between the words. She knew a great deal about seeing. Asked to let her mother see something, she held it up and away from her body. Asked to make it so her mother could not see it, she hid it behind her. What the relations could not give her was the seeing itself.

That is the first limit of the relations between words. They carry the structure of an experience, not the experience. A machine trained only

on text can know that red sits between orange and purple, that it is the colour of blood and of stop signs, that it is warm and urgent. It has never seen red.

The second limit is that relations between words can tell you that a sentence fits. They cannot tell you that it is true. Kober and Ventris had the tripod tablet. The machine often has nothing like it. A made-up court ruling, with a plausible name, a plausible year and a plausible court, fits the network of legal language perfectly. It never happened. The machine's most famous failure, its habit of inventing facts with complete confidence, follows directly from this. It is what you would expect from something that learned the shape of our sentences so well that it can produce the shape without the substance.

We wrote down, endlessly, how a true sentence sounds. We almost never wrote down how we checked it. The proof in a textbook appears polished, without the twenty attempts that failed. And so a machine that learned from our writing learned to sound right far more reliably than it learned to be right.

Which raises a question that will follow us through every chapter: **who checks?** When the machine's words are measured against something, what is that something? At first it is nothing but more text. Later it becomes human taste, then the feelings of the person typing, and only sometimes the world itself. The answer, I will argue, leads to the most important rule in this book: the machine is good at exactly what its judge can check, and it learns to please its judge even when the judge is wrong.

THE MAP OF THIS BOOK

The first of the book's five parts, *The Archive*, asks how human thought got into words in the first place, and why it got there implicitly, in the relations between words, rather than as anything anyone wrote down. It tells why decades of trying to write meaning down as rules failed.

The second, *The Machine of Relations*, opens the machine. It tells the history of a rival dream, machines that learn from examples instead of rules, from the first crude classifiers to programs that choose the next word; it shows how a vast web of adjustable numbers, which nobody programmed line by line, came to hold meaning; why guessing the next word forced it to learn the network of relations; what that network looks like inside; and how it compares with the other great web of connections that holds knowledge without writing it down, the human brain.

The third, *The Reader*, follows the machine as it reads our archive. It shows what the network of relations preserves and what it loses, and how not only our knowledge but our ways of reasoning, and even our many voices, turned out to be patterns between the words.

The fourth, *The Mirror*, turns to what the machine does to us when nothing anchors its words: the confident inventions, the flattery, the strange gap between how productive it makes us feel and how productive it makes us, whose words the network is made of, and the unsettling fact that nobody, including the machine, can read more than a fraction of what is inside it.

The fifth, *The Judge*, asks what happens when the machine is finally checked against the world: when it learns from experiments rather than books, when its words become actions, when it learns to fool its judges, and what that leaves for us to do.

* * *

Here is the first rung of the ladder this book will climb. Knowledge can exist without ever being written down anywhere. Kober and Ventris found a language in the relations among signs. A machine found much of what we know in the relations among our words. If that sounds strange, think of how you learned most of what you know. Very little of it was ever defined for you. Like Kelli, you absorbed it from the way people talk.

A question to take with you: how much of what you know did anyone ever actually tell you, and how much did you pick up from the way people talk?

If knowledge can live between the words, how did it get there in the first place? To answer that, we have to start long before the machine, with the first people who learned to keep their thoughts outside their heads.

* * *

© 2026 Yechiel B. All rights reserved. A draft from a book in progress.

NOTES

1. On the script, the find-spots and the state of knowledge before 1950, see the University of Cambridge account of the decipherment (Mycenaean Epigraphy Group) and John Chadwick, *The Decipherment of Linear B* (Cambridge University Press, 1958). The tablets are usually dated between about 1400 and 1200 BCE; the dating of the Knossos tablets is debated.
2. Kober's papers, including her card index and two cigarette cartons (a Lucky Strike and a Chesterfield) in which she filed the cards, are held by the Program in Aegean Scripts and Prehistory (PASP), University of Texas at Austin. The archive counts "over 180,000 items" in the cigarette-carton and other files. Paper was scarce in wartime, so Kober cut the cards by hand from scrap: exam-book covers, old greeting cards and library checkout slips (Margalit Fox, *The Riddle of the Labyrinth*, 2013; Fox, interview on NPR, 30 June 2013). Popular accounts give 180,000 or 186,000 cards.
3. Alice E. Kober, "Inflection in Linear Class B: I – Declension," *American Journal of Archaeology* 50, no. 2 (1946): 268–276 (printed 1947); also her papers in *AJA* 49 (1945) and 52 (1948). Kober worked with the signs themselves, before any had a sound value. The word-sets quoted in popular accounts ("Kober's triplets", such as *ko-no-si-ja* / *ko-no-si-jo* / *ko-no-so*) are given in Ventris's later readings and should not be read as what Kober saw. In such sets the shortest form drops a sign and changes the one before it, which is why the text says the forms differed in their endings. See also Davis and Bronzo, "Deciphering Kober's Contribution to the Decipherment of Linear B," *SMEA* NS 9 (2023): 83–98.
4. Kober was born in New York on 23 December 1906 and taught Classics at Brooklyn College from 1930 until her death. Her cause of death is not recorded in her death certificate or obituaries; the book does not state it.
5. Ventris was born on 12 July 1922. In 1936, as a pupil at Stowe School, he met Evans, then eighty-five, at an exhibition at Burlington House in London. Accounts differ on whether Evans was lecturing or guiding a tour, and the words Ventris is supposed to have said to him come from later recollection. See Andrew Robinson, *The Man Who Deciphered Linear B* (2002).
6. The grid illustrated in the Cambridge account is from Ventris's Work Note 17 (February 1952). Ventris's earlier "Mid-Century Report" (1950) surveyed the views of other scholars.
7. *a-mi-ni-so* (Amnisos) fixed two sign values; the grid then turned another word into *ko-no-so* (Knossos). Ventris had earlier argued, in his first article (1940), for a link with Etruscan.
8. Work Note 20 (1 June 1952) first raised the possibility that the tablets were Greek. The BBC Third Programme broadcast was on 1 July 1952. John Chadwick, who heard it, was the first to write to Ventris; their joint paper, "Evidence for Greek Dialect in the Mycenaean Archives," appeared in *Journal of Hellenic Studies* 73 (1953), and their *Documents in Mycenaean Greek* in 1956.
9. The tablet is PY Ta 641, from the Palace of Nestor at Pylos, found in two pieces on 4 and 10 June 1952 and legible only after cleaning in spring 1953. It reads *ti-ri-po-de* beside tripod ideograms, *qe-to-ro-we* (four-handled), *ti-ri-o-we-e* and *ti-ri-jo-we* (three-handled) and *a-*

no-we (without handles) beside matching drawings of vessels. Blegen's letter to Ventris of 16 May 1953 is quoted in Chadwick 1958 and dated in Stephen V. Tracy, "The Acceptance of the Greek Solution for Linear B," *Hesperia* 87 (2018): 1–16.

10. Late at night, driving home from his in-laws' house, he struck a lorry parked at the roadside near Hatfield, Hertfordshire, and was killed instantly; the coroner's verdict was accidental death (Andrew Robinson, *The Man Who Deciphered Linear B*, 2002, 150–152). Speculation that it was not an accident is unsupported, and the book does not repeat it.

11. On the honeybee dance: Karl von Frisch, who shared the 1973 Nobel Prize in Physiology or Medicine for this work. On vervet alarm calls: Robert Seyfarth, Dorothy Cheney and Peter Marler, "Monkey Responses to Three Different Alarm Calls," *Science* 210 (1980): 801–803.

12. That no human community has ever been found without a full language, and that no language is "primitive" in its grammar, is a standard observation in linguistics. See, for example, Steven Pinker, *The Language Instinct* (1994), chapter 2, "Chatterboxes", which notes that no mute community has ever been found and calls the universality of complex language a discovery that awes linguists.

13. René Descartes, *Discourse on the Method* (1637), Part V. Paraphrased.

14. A. M. Turing, "Computing Machinery and Intelligence," *Mind* 59, no. 236 (1950): 433–460. Turing set aside the question "Can machines think?" and replaced it with the "imitation game". His prediction (p. 442) was that in about fifty years an average interrogator would have no more than a 70 per cent chance of a right identification after five minutes of questioning; he called it a conjecture. The often-quoted "30 per cent pass mark" was read into this later. The paper does not cite Descartes, but Turing marked a Descartes passage about a parrot in his copy of Geoffrey Jefferson's 1949 Lister Oration (King's College archive, reported by Gonçalves, *IEEE Annals of the History of Computing*, 2024).

15. OpenAI, "Introducing ChatGPT," 30 November 2022, https://openai.com/index/chatgpt/

16. Sam Altman, post on X (then Twitter), 5 December 2022.

17. As of autumn 2026. OpenAI stated that ChatGPT had "more than 1 billion weekly active users" ("A milestone in expanding access to AI," 31 August 2026). Google's chief executive Sundar Pichai announced on 11 August 2026 that the Gemini app had passed a billion monthly users. These are company claims, and companies count users differently (weekly against monthly, one app against many).

18. Aaron Chatterji, Tom Cunningham, David Deming, Zoë Hitzig, Christopher Ong, Carl Shan and Kevin Wadman, "How People Use ChatGPT," NBER Working Paper 34255 (September 2025). Based on about 1.1 million conversations from consumer accounts, classified automatically: 49% "asking", 40% "doing", 11% "expressing"; by mid-2025 about 73% of messages were not related to work. Two of the authors are OpenAI employees, and business accounts are excluded.

19. Emily M. Bender, Timnit Gebru, Angelina McMillan-Major and Margaret Mitchell, "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?" *FAccT '21* (March 2021), https://dl.acm.org/doi/10.1145/3442188.3445922

20. International Mathematical Olympiad, July 2025: an advanced version of Google DeepMind's Gemini Deep Think scored 35 of 42, gold-medal level, officially graded by IMO coordinators and working end to end in natural language within the time limit (Google DeepMind, 21 July 2025). OpenAI reported the same score for an experimental model, graded by former medallists rather than by the IMO. On software flaws, see Chapter 17.

21. This describes pretraining, the first and by far the largest stage of training. Later stages shape the model with examples of good answers and with feedback (Chapters 12 and 16), but they too supply examples and judgments, not rules of grammar or lists of facts. Meta's Llama 3, for example, was trained on about 15 trillion tokens (word-pieces).

22. On "reporting bias", the tendency of text to mention the unusual rather than the typical, see Jonathan Gordon and Benjamin Van Durme, "Reporting Bias and Knowledge Acquisition," *Proceedings of the 2013 Workshop on Automated Knowledge Base Construction* (ACM, 2013): 25–30. That text-only models nonetheless recover perceptual structure such as the arrangement of colours is shown in Chapter 10. The banana example is an illustration, not a measured case.

23. The idea that a word's meaning lies in its relations to other words is old. The linguists J. R. Firth and Zellig Harris stated versions of it in the 1950s, and it is the working assumption of modern word vectors. Several recent writers have argued that it explains what language models learned, among them Steven Piantadosi and Felix Hill ("Meaning without Reference in Large Language Models," 2022), Christopher Summerfield (*These Strange New Minds*, 2025) and Blaise Agüera y Arcas (*What Is Intelligence?*, 2025). This book does not claim the idea as new. What it adds is to follow it through, and to ask what the relations leave out.

24. Harry Beck's 1933 diagram of the London Underground straightened the lines, evened out the spacing between stations and enlarged the centre while compressing the outer branches, so distances on it are distorted. Passengers trust it anyway: a study of London Underground journeys found that route choices respond about twice as strongly to distance on the map as to actual travel time (Zhan Guo, "Mind the map! The impact of transit maps on path choice in public transit," *Transportation Research Part A* 45, no. 7, 2011: 625–639). A 2025 study recovered a faint, statistically detectable trace of the layout of underground stations in European cities from word vectors alone (for distances, in four of the five cities studied), though the effect is small: Anceresi, Gatti, Vecchi, Marelli and Rinaldi, "A map of words: Retrieving the spatial layout of medium-scale geographical maps through distributional semantics," *Neuropsychologia* 217 (2025): 109190.

25. Barbara Landau and Lila R. Gleitman, *Language and Experience: Evidence from the Blind Child* (Harvard University Press, 1985); Lila Gleitman, "The Structural Sources of Verb Meanings," *Language Acquisition* 1 (1990): 3–55, which reproduces the "look up" drawings. The experiments were run with Kelli at around three years old (later papers say four); three blind children were followed, but the experiments described here were with Kelli alone. The researchers concluded that the syntax of her mother's speech, more than the situations she was in, distinguished *look* and *see* from other verbs. Spelled "Kelli" by Landau (2021) and "Kelly" by Bedny et al. (2019). Modern follow-up: congenitally blind and sighted adults judge the meanings of vision verbs almost identically (Bedny et al., *Cognition* 2019).